

Data Intelligence Platform - Analytics

INTRODUCTION TO DATABRICKS



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Data Practitioner

Why do organizations care about analytics?



Supported Languages

Scala

- Based on Java
- Generally used for data engineering

Python

- Used for all use cases

SQL

- Used for data engineering and BI

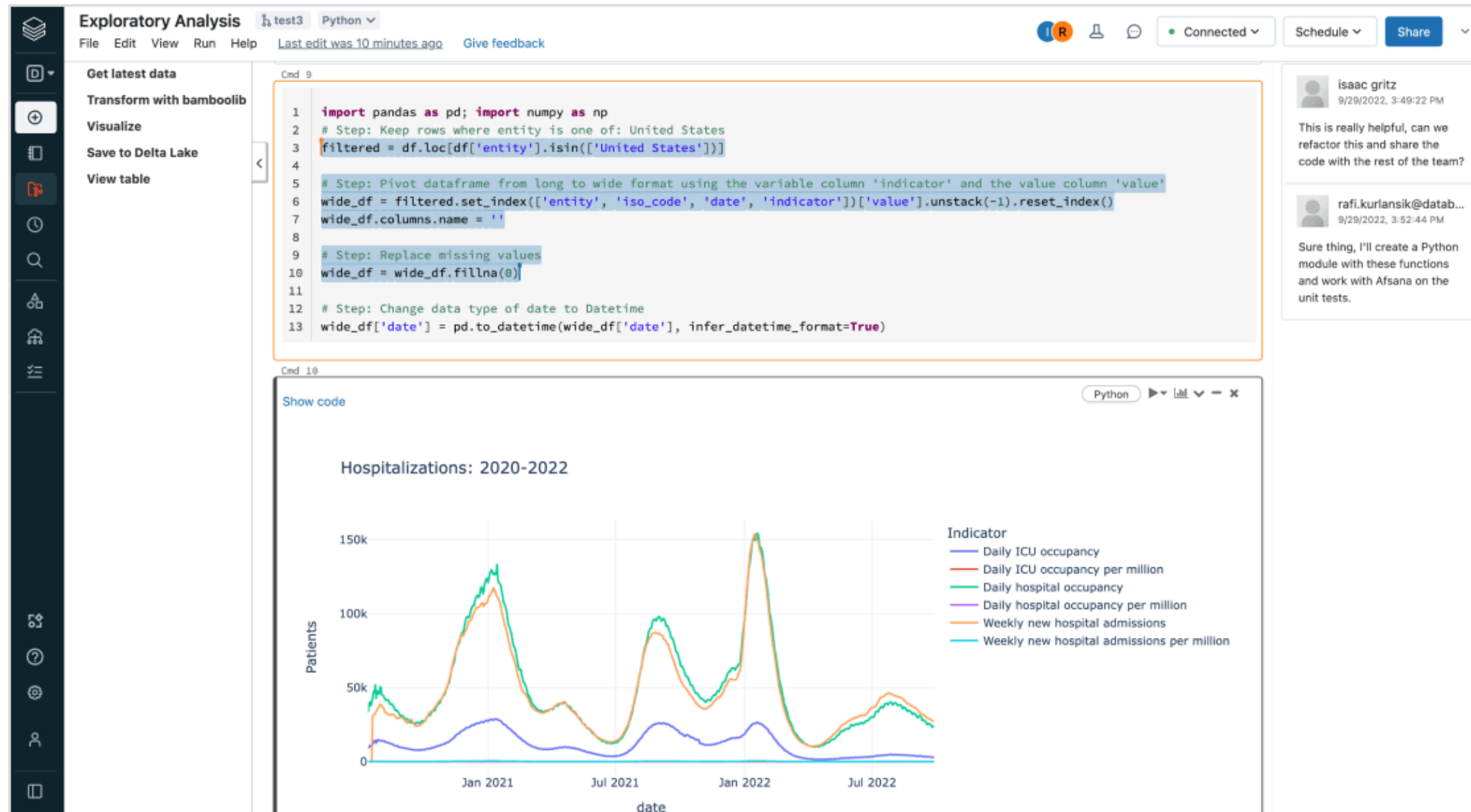
R

- Used for data science use cases



Databricks Notebooks

Based on Jupyter notebooks, Databricks Notebooks are an optimized and enhanced version.



SQL Editor

The screenshot displays the Databricks SQL Editor interface. On the left is a dark sidebar with navigation icons. The main area is divided into three panels:

- Catalog:** A sidebar on the left with a search bar "Type to filter" and a list of databases: `hive_metastore`, `ml`, `samples`, and `system`.
- Query Editor:** The central panel showing a SQL query. The query is:

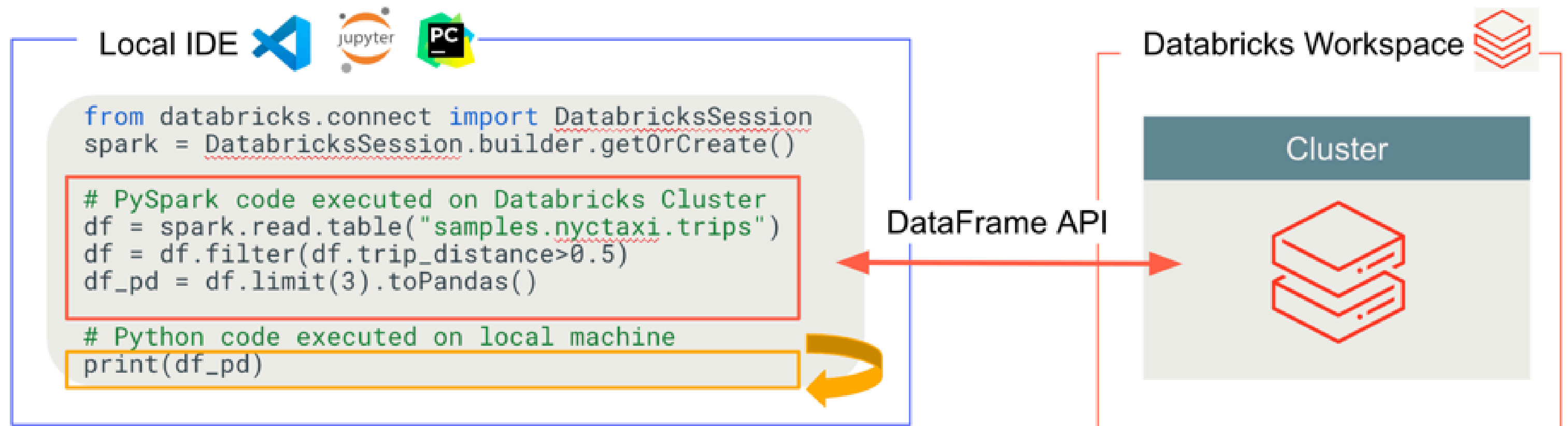
```
1 SELECT
2   o_orderdate AS Date,
3   o_orderpriority AS Priority,
4   sum(o_totalprice) AS `Total Price`
5 FROM
6   `samples`.`tpch`.`orders`
7 WHERE
8   o_orderdate > '1994-01-01'
9   AND o_orderdate < '1994-01-31'
10 GROUP BY
11   1,
12   2
13 ORDER BY
14   1,
15   2
```
- Results:** The bottom panel showing the query results in a table. The table has columns: `Date`, `Priority`, and `Total Price`. The results are 145 rows, showing dates from 1994-01-02 to 1994-01-03 and various priorities.

At the bottom of the interface, there is a status bar indicating "12 s 751 ms | 145 rows returned" and "Refreshed a minute ago".

¹ <https://docs.databricks.com/en/sql/user/sql-editor/index.html>

Databricks Connect

Bring your own IDE! Leverage the power of Databricks Clusters while working in your favorite coding environment.



¹ <https://docs.databricks.com/en/dev-tools/databricks-connect/index.html>

Let's practice!
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SQL in the Data Intelligence Platform

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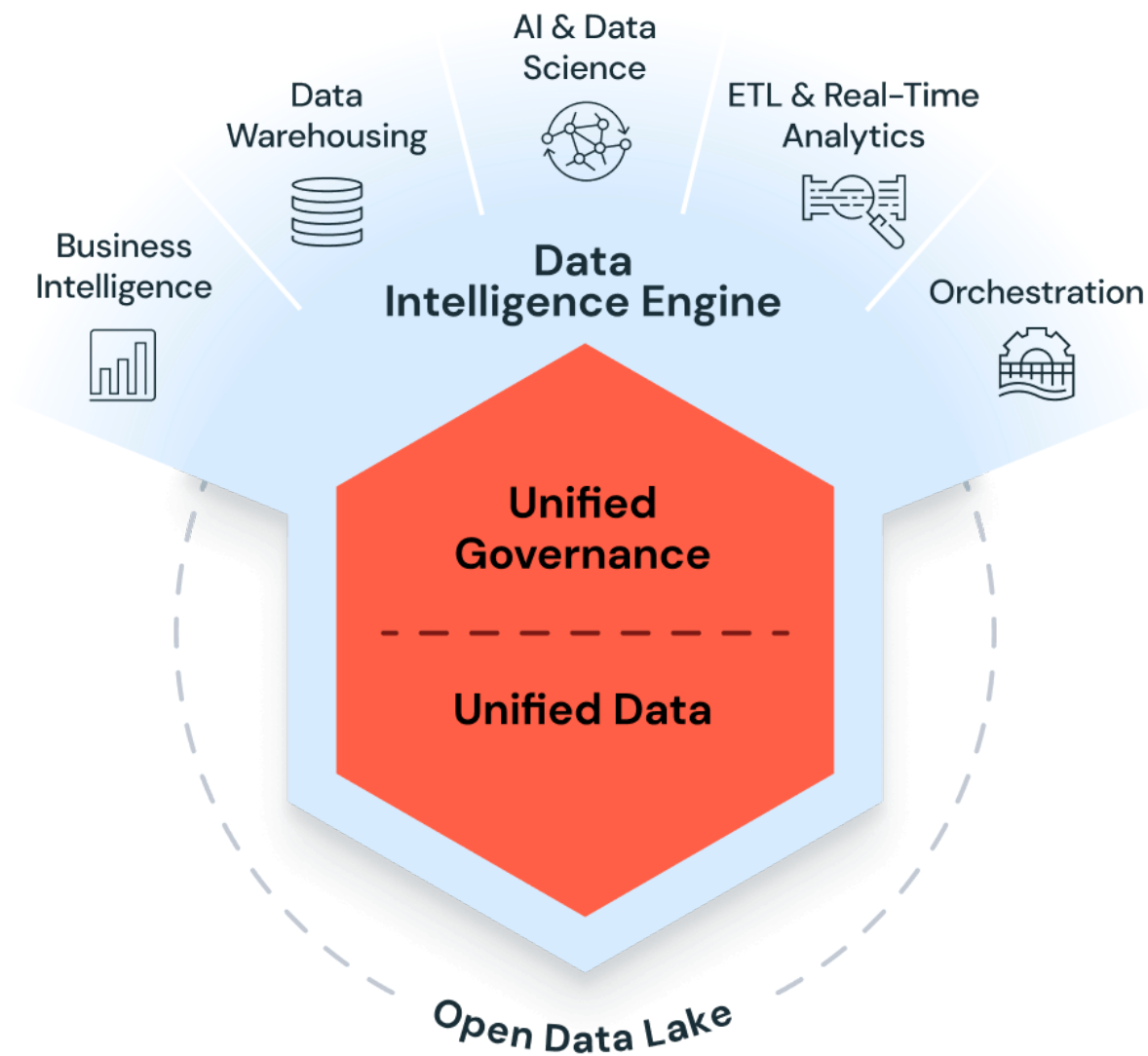
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Databricks SQL

The Data Intelligence Platform is designed to run your most challenging data warehousing workloads.



Databricks for SQL Users

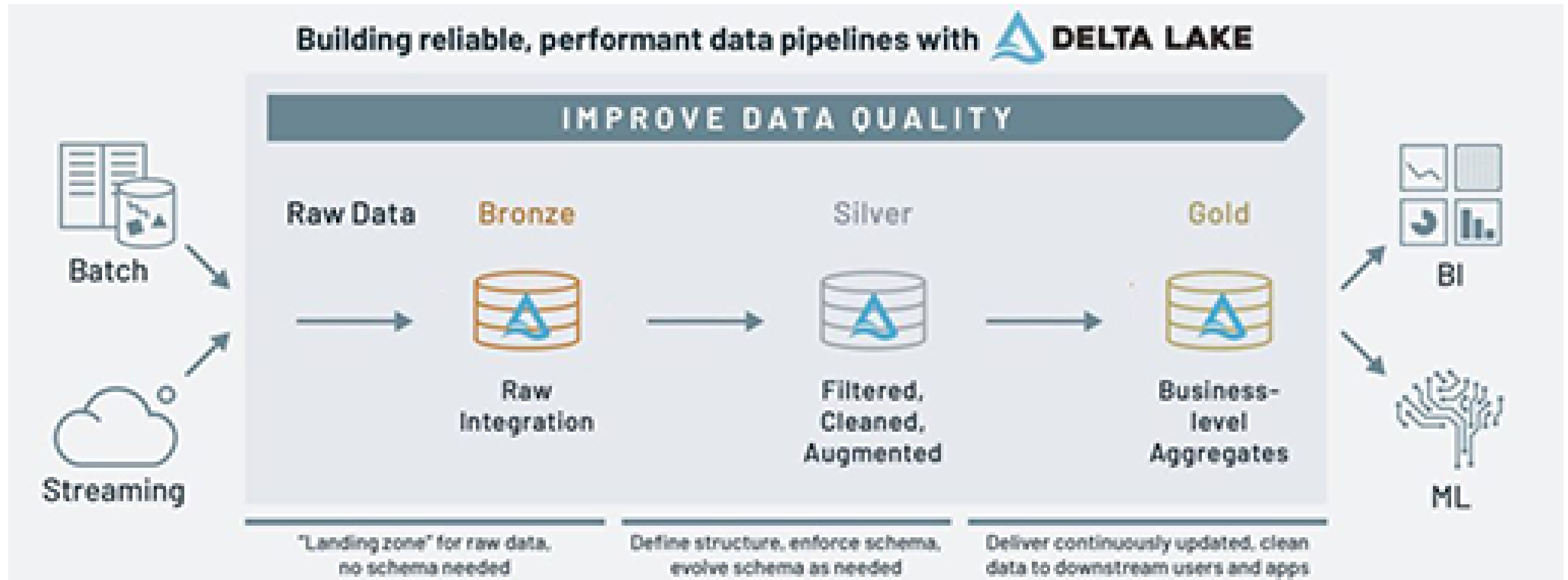


Databricks SQL

- Data Warehousing for the Lakehouse
- Familiar environment for SQL users
- SQL-optimized performance (Photon)
- Connect to your favorite BI tools

Comes built into the platform!

SQL in the Data Intelligence Platform



SQL in the Data Intelligence Platform

Benefits

- Fully integrated
- Scalable, performant compute
- Great UI for analysts

Key Features

- ANSI SQL
- Enhanced Photon engine
- Built-in visualizations

```
SELECT *  
FROM json.`/Volumes/  
        catalog_name/schema_name/  
        volume_name/path/to/data`  
  
SELECT *  
FROM catalog_name.schema_name.table_name  
  
CREATE TABLE  
        catalog_name.schema_name.table_name  
  
AS  
  
SELECT *  
FROM ...
```

Let's review!

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Congratulations!

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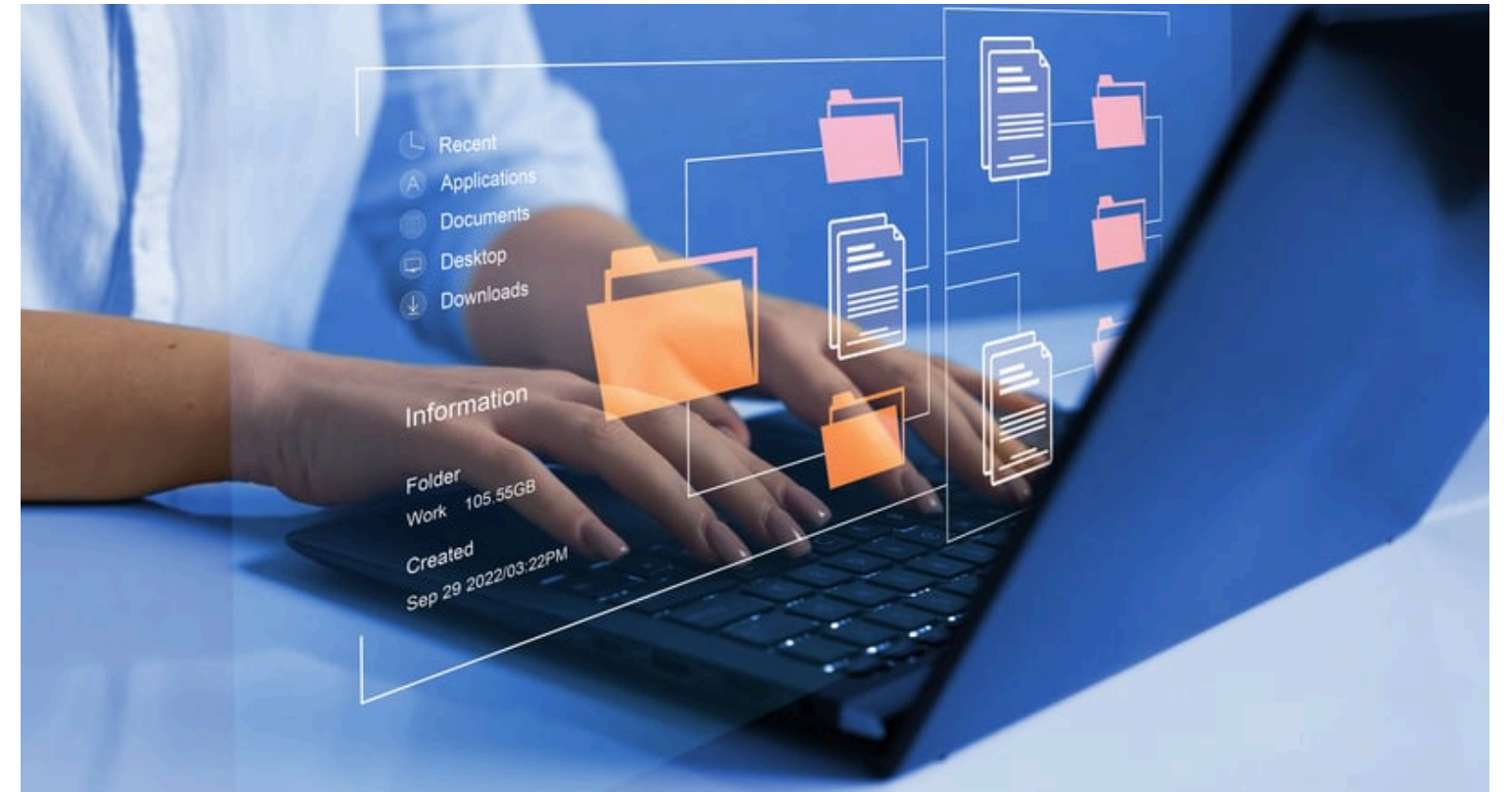
Chapter 1

- Introduction to the Data Intelligence Platform
- Key benefits of building on the platform
- Understanding the underlying architecture



Chapter 2

- Learning about Unity Catalog and Delta
- Managing organizational data
- Utilizing clusters and SQL Warehouses for scalable compute power

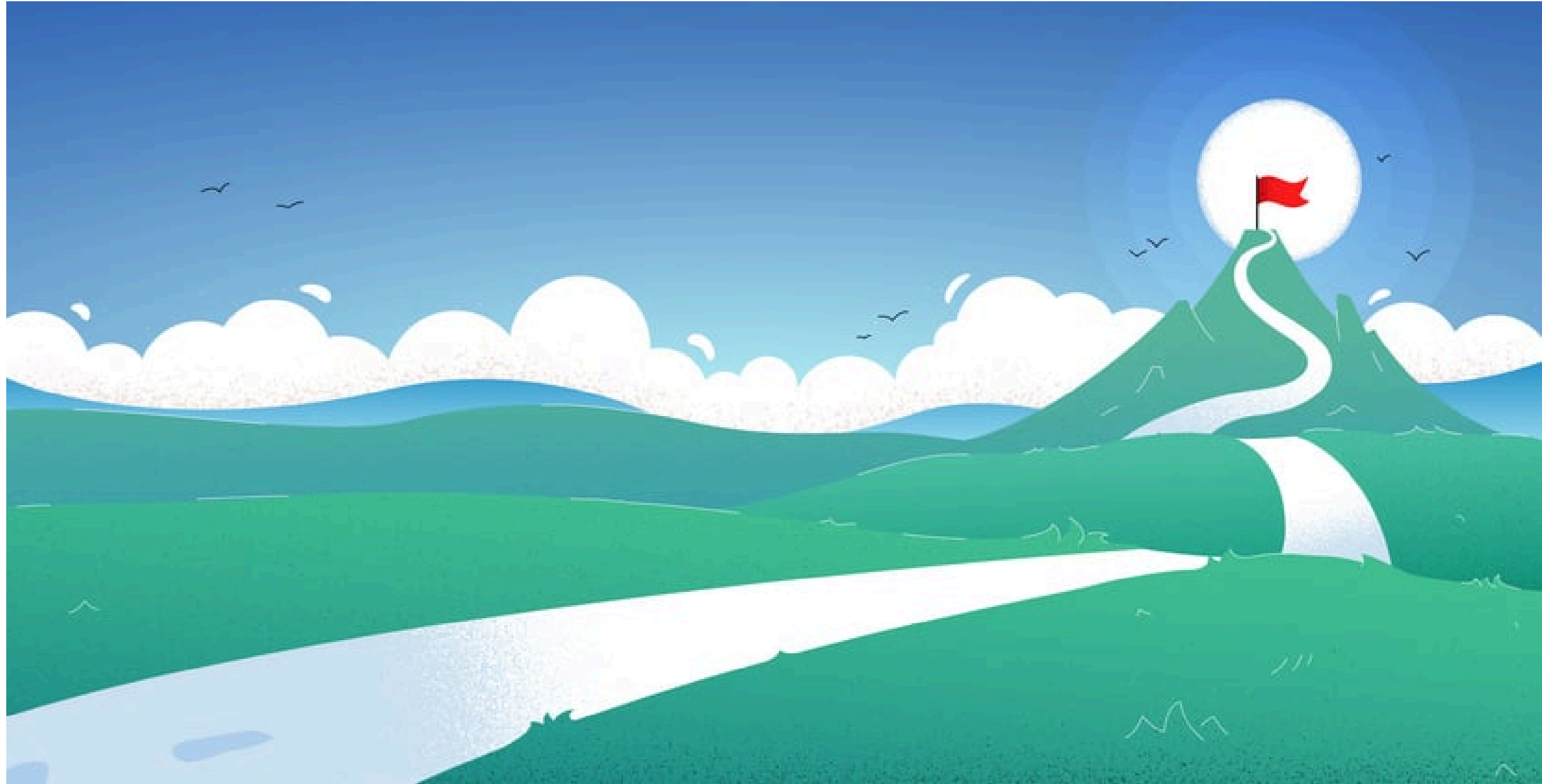


Chapter 3

- Conducting analytical processes in Databricks
- Using Databricks as an enterprise Data Warehouse
- Exploring Databricks SQL for analytics



Just the start



Congratulations!

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